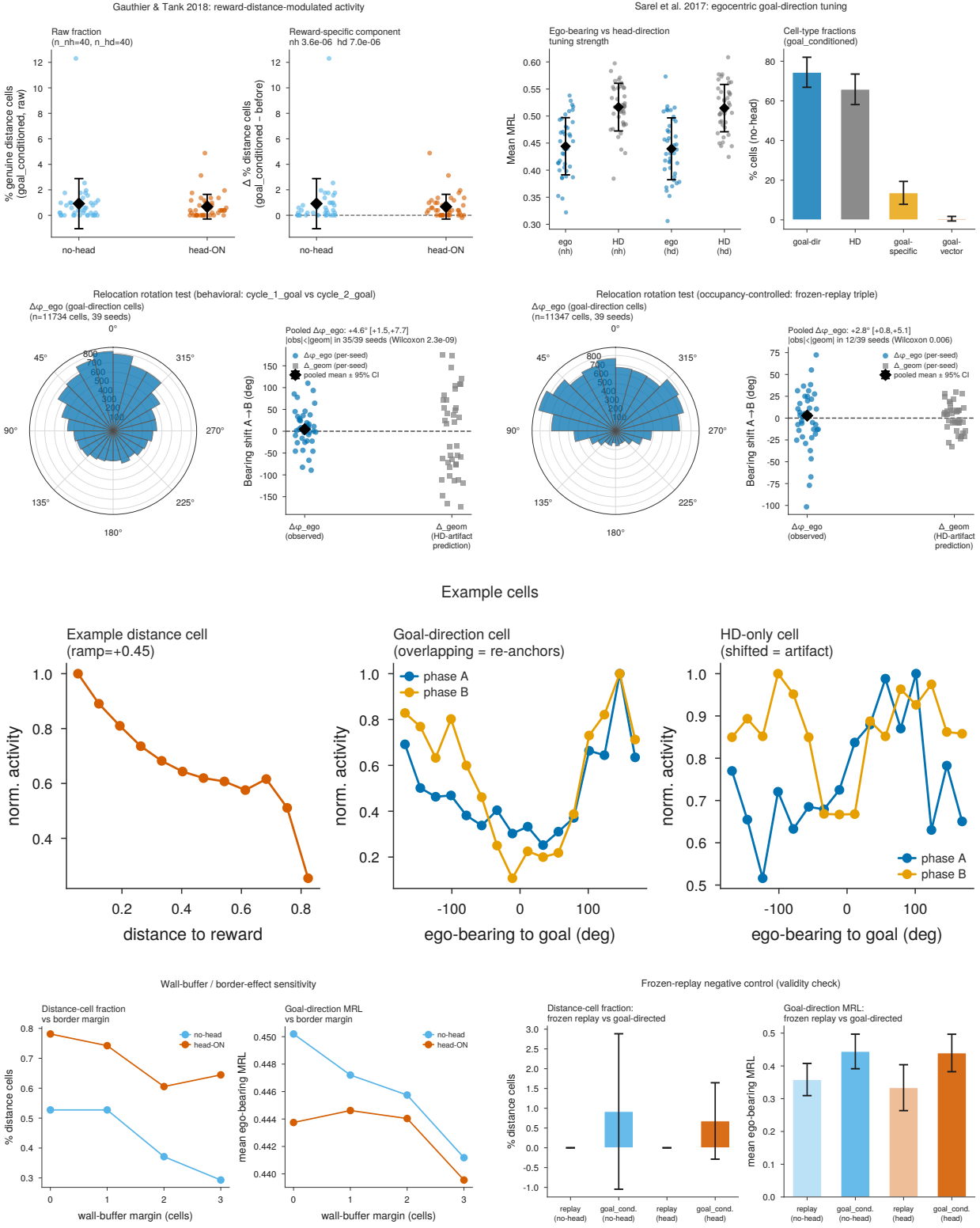

Gauthier & Tank 2018: reward-distance-modulated activity
Raw fraction
(n_nh=40, n_hd=40)
% genuine distance cells
(goal_conditioned, raw)
no-head
head-ON
Reward-specific component
nh 3.6e-06 hd 7.0e-06
Δ % distance cells
(goal_conditioned − before)
Sarel et al. 2017: egocentric goal-direction tuning
Ego-bearing vs head-direction
tuning strength
Mean MRL
ego (nh)
HD (nh)
ego (hd)
HD (hd)
Cell-type fractions
(goal_conditioned)
% cells (no-head)
goal-dir
HD
goal-specific
goal-vector
Relocation rotation test (behavioral: cycle_1_goal vs cycle_2_goal)
Δφ_ego (goal-direction cells)
(n=11734 cells, 39 seeds)
Pooled Δφ_ego: +4.6° [+1.5,+7.7]
|obs|<|geom| in 35/39 seeds (Wilcoxon 2.3e-09)
Δφ_ego (per-seed)
Δ_geom (per-seed)
pooled mean ± 95% CI
Bearing shift A→B (deg)
Δφ_ego
(observed)
Δ_geom
(HD-artifact
prediction)
Relocation rotation test (occupancy-controlled: frozen-replay triple)
Δφ_ego (goal-direction cells)
(n=11347 cells, 39 seeds)
Pooled Δφ_ego: +2.8° [+0.8,+5.1]
|obs|<|geom| in 12/39 seeds (Wilcoxon 0.006)
Example cells
Example distance cell
(ramp=+0.45)
norm. activity
distance to reward
Goal-direction cell
(overlapping = re-anchors)
phase A
phase B
ego-bearing to goal (deg)
HD-only cell
(shifted = artifact)
Wall-buffer / border-effect sensitivity
Distance-cell fraction
vs border margin
no-head
head-ON
% distance cells
wall-buffer margin (cells)
Goal-direction MRL
vs border margin
mean ego-bearing MRL
Frozen-replay negative control (validity check)
Distance-cell fraction:
frozen replay vs goal-directed
replay
(no-head)
goal_cond.
(no-head)
replay
(head)
goal_cond.
(head)
Goal-direction MRL:
frozen replay vs goal-directed